Lab Assets (Runnable Classroom Kits)

Notes teach; **these folders run**. Start every major concept with a **customer ticket**, then `starter/` or `broken/`.

Layout convention

```
15_Lab_Assets/
  <nn>_<name>/
    README.md     # ticket (start here)
    starter/      # incomplete — learners extend
    broken/       # intentional bugs — debug lab
    solution/     # reference answers (do not share; pack script excludes)
    seed/         # sample CSV/JSON/SQL when needed
```

Full index (all concepts)

Kit	Sessions	Ticket
01_python_expense_tracker	~42–45	Ticket: Expense Tracker CSV Export
02_mysql_library_seed	~50–55	Ticket: Library join report
03_django_auth_tasks	~71–77	Ticket: Private tasks per user
04_drf_products	~82–85	Ticket: Product JSON API for mobile
06_cli_calculator	~06–10	Ticket: Campus Canteen Calculator
07_casting_bugs	~09	Ticket: Marks Entry Keeps Crashing
08_atm_menu	~11–15	Ticket: Mini ATM Menu
09_string_parser	~16–17	Ticket: Registration Field Cleaner
10_marks_manager	~18–20	Ticket: Student Marks Manager (CLI)
11_contact_book	~23–25	Ticket: Contact Book
12_structure_chooser	~26–27	Ticket: Pick the Right Structure
13_function_refactor	~28–32	Ticket: Refactor ATM Into Functions
14_decorator_login_gate	~30–31	Ticket: Login Gate Decorator
15_generator_csv_stream	~31	Ticket: Stream Large CSV
16_venv_package_bootstrap	~33–35	Ticket: Ship a Tiny Utils Package
17_bank_account_oop	~36–39	Ticket: BankAccount Class
18_safe_input_exceptions	~41	Ticket: Never Crash on Bad Input
19_git_portfolio_hygiene	~44–45	Pointer → `GIT/labs/01_git_portfolio_hygiene`
20_sql_injection_demo	~55	Ticket: Stop SQL Injection

Kit	Sessions	Ticket
24_django_url_views_stub	~63–67	Ticket: Two Pages With One Base Template
25_flask_notes_api	~81	Ticket: Tiny Notes JSON API
26_unittest_utils	~91–92	Ticket: Tests Must Catch the Bug
27_terminal_workspace	~01–02	Ticket: Messy Desktop → Clean Workspace
29_ai_review_stub	optional after S95	Ticket: Optional AI Recommend Endpoint
30_ksr_utils_toolkit	~28–35	Ticket: Ship `ksr_utils` (validators, formatters, decorators, generators)
31_email_id_dedup	~21–22	Ticket: Clean Duplicate IDs Before Import
32_campus_wallet_abc	~40	Ticket: HDFC Bank Campus Wallet

Shared topic labs (outside this folder)

Kit	Location
Git / GitHub labs	`GIT/labs/`
HTML / CSS / JS labs	`FRONTEND/labs/`
React labs	`FRONTEND/REACT/labs/`

How to use a kit

1. Open the kit folder (or copy `starter/` / `broken/` into your workspace).
2. Read the kit `README.md` ticket first — code before long theory.
3. Work the ticket solo or in pairs.
4. Compare with `solution/` only after you finish (or when told in class).

Student pack policy

`solution/` folders are **excluded** by `scripts/pack_student_share.sh` and should not be shared. Starter/broken/seed remain.

Map to modules

Lab kits live under this folder. Session order stays in Session_Calendar_100.md. Modules link kits from **Mini-Project** (and practice) — there is no separate Classroom Ticket section.